

A Technical Introduction to Stable Diffusion

$$\mathcal{L} = \mathbb{E}_{x, \epsilon, t} [\|\epsilon - \epsilon_{\theta}(x_t, t)\|^2]$$

Section 1

Development Stable Diffusion originated from a project called Latent Diffusion, developed in Germany by researchers at LMU Munich in Munich and Heidelberg University. Four of the original 5 authors (Robin Rombach, Andreas Blattmann, Patrick Esser and Dominik Lorenz) later joined Stability AI and released subsequent versions of Stable Diffusion. The technical license for the model was released by the CompVis group at LMU Munich. Development was led by Patrick Esser of Runway and Robin Rombach of CompVis, who were among the researchers who had earlier invented the latent diffusion model architecture used by Stable Diffusion. Stability AI also credited EleutherAI and LAION (a German nonprofit which assembled the dataset on which Stable Diffusion was trained) as supporters of the project.

Section 2

User interfaces Stability provides an online image generation service called DreamStudio. The company also released an open source version of DreamStudio called StableStudio. In addition to Stability's interfaces, many third party open source interfaces exist, such as AUTOMATIC1111 Stable Diffusion Web UI, which is the most popular and offers extra features, Fooocus, which aims to decrease the amount of prompting needed by the user, and ComfyUI, which has a node-based user interface, essentially a visual programming language akin to many 3D modeling applications.

Section 3

Getty Images asserted that the training and development of Stable Diffusion involved the unauthorized use of its images, which were downloaded on servers and computers that were potentially in the UK. However, Stability AI argued that all training and development took place outside the UK, specifically in U.S. data centers operated by Amazon Web Services. Stability AI applied for reverse summary judgment and/or strike out of two claims: the training and development claim, and the secondary infringement of copyright claim. The High Court, however, refused to strike out these claims, allowing them to proceed to trial. The court is to determine whether the training and development of Stable Diffusion occurred in the UK, which is crucial for establishing jurisdiction under the UK's Copyright, Designs and Patents Act 1988 (CDPA). The

secondary infringement claim revolved around whether the pre-trained Stable Diffusion software, made available in the UK through platforms like GitHub, HuggingFace, and DreamStudio, constitutes an "article" under sections 22 and 23 of the CDPA. As of November 4, 2025 Getty Images largely lost the lawsuit over AI image generator to Stability AI, prompting Getty and some lawyers to call for stronger protections for copyright owners in Britain.

Section 4

The 3.0 version completely changes the backbone. Not a UNet, but a Rectified Flow Transformer, which implements the rectified flow method with a Transformer. The Transformer architecture used for SD 3.0 has three "tracks", for original text encoding, transformed text encoding, and image encoding (in latent space). The transformed text encoding and image encoding are mixed during each transformer block. The architecture is named "multimodal diffusion transformer (MMDiT), where the "multimodal" means that it mixes text and image encodings inside its operations. This differs from previous versions of DiT, where the text encoding affects the image encoding, but not vice versa.

Section 5

An "embedding" can be trained from a collection of user-provided images, and allows the model to generate visually similar images whenever the name of the embedding is used within a generation prompt. Embeddings are based on the "textual inversion" concept developed by researchers from Tel Aviv University in 2022 with support from Nvidia, where vector representations for specific tokens used by the model's text encoder are linked to new pseudo-words. Embeddings can be used to reduce biases within the original model, or mimic visual styles. A "hypernetwork" is a small pretrained neural network that is applied to various points within a larger neural network, and refers to the technique created by NovelAI developer Kurumuz in 2021, originally intended for text-generation transformer models. Hypernetworks steer results towards a particular direction, allowing Stable Diffusion-based models to imitate the art style of specific artists, even if the artist is not recognised by the original model; they process the image by finding key areas of importance such as hair and eyes, and then patch these areas in secondary latent space. DreamBooth is a deep learning generation model developed by researchers from Google Research and Boston University in 2022 which can fine-tune the model to generate precise, personalised outputs that depict a specific subject, following training via a set of images which depict the subject.

Section 6

Stable Diffusion is a deep learning, text-to-image model released in 2022 based on diffusion techniques. The

generative artificial intelligence technology is the premier product of Stability AI and is considered to be a part of the ongoing AI boom. It is primarily used to generate detailed images conditioned on text descriptions, though it can also be applied to other tasks such as inpainting, outpainting, and generating image-to-image translations guided by a text prompt. Its development involved researchers from the CompVis Group at LMU Munich and Runway with a computational donation from Stability and training data from non-profit organizations. Stable Diffusion is a latent diffusion model, a kind of deep generative artificial neural network. Its code and model weights have been released publicly, and an optimized version can run on most consumer hardware equipped with a modest GPU with as little as 2.4 GB VRAM. This marked a departure from previous proprietary text-to-image models such as DALL-E and Midjourney which were accessible only via cloud services.

Section 7

Training data Stable Diffusion was trained on pairs of images and captions taken from LAION-5B, a publicly available dataset derived from Common Crawl data scraped from the web, where 5 billion image-text pairs were classified based on language and filtered into separate datasets by resolution, a predicted likelihood of containing a watermark, and predicted "aesthetic" score (e.g. subjective visual quality). The dataset was created by LAION, a German non-profit which receives funding from Stability AI. The Stable Diffusion model was trained on three subsets of LAION-5B: laion2B-en, laion-high-resolution, and laion-aesthetics v2 5+. A third-party analysis of the model's training data identified that out of a smaller subset of 12 million images taken from the original wider dataset used, approximately 47% of the sample size of images came from 100 different domains, with Pinterest taking up 8.5% of the subset, followed by websites such as WordPress, Blogspot, Flickr, DeviantArt and Wikimedia Commons. An investigation by Bayerischer Rundfunk showed that LAION's datasets, hosted on Hugging Face, contain large amounts of private and sensitive data.